

Analytic Cut in Epistemic Logics with Distributed Knowledge

Ryo Murai Sizhuo Liu¹ Katsuhiko Sano

Hokkaido University

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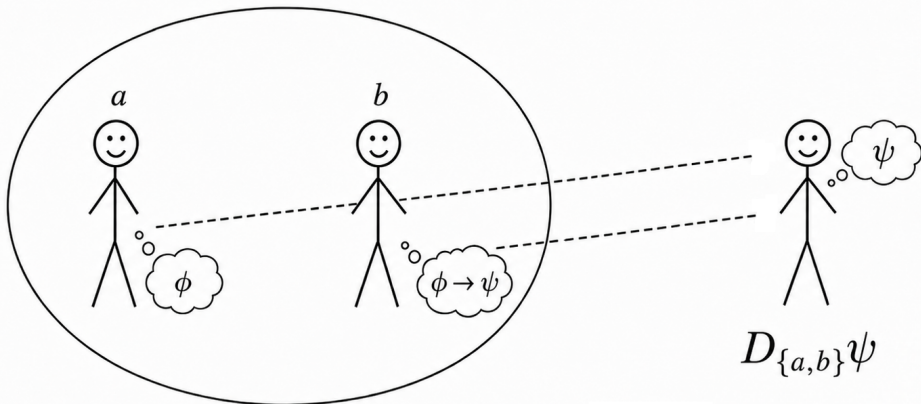
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Motivation

Sequent Calculi with D_G

Cut-elimination holds in $G(\mathbf{K}_D)$, $G(\mathbf{KT}_D)$, $G(\mathbf{S4}_D)$, etc. but **not** in $G(\mathbf{S5}_D)$ (Murai & Sano 2020).

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Agent Interpolation for Logics with D_G

The condition of **agents** has only recently been discussed explicitly (Murai & Sano 2020; Umemura & Sano 2025; Bílková et al. 2026).

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What we found

Although $G(\mathbf{K45})$ and $G(\mathbf{KD45})$ are cut-free, $G(\mathbf{K45}_D)$ and $G(\mathbf{KD45}_D)$ w/ D_G s are **not**.

What's New on This Paper?

Main Results

- 1 $G(\mathbf{K45}_D)$, $G(\mathbf{KD45}_D)$ and $G(\mathbf{S5}_D)$:
not cut-free, but enjoys the subformula property (i.e., analytic) as in (Takano 1992).

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- 1 $G(\mathbf{K45}_D)$, $G(\mathbf{KD45}_D)$ and $G(\mathbf{S5}_D)$:
not cut-free, but enjoys the subformula property (i.e., analytic) as in (Takano 1992).
- 2 Craig Interpolation of all the calculi above:
if $G(\Lambda) \vdash \Rightarrow \varphi \rightarrow \psi$,
then there exists an interpolant χ s.t. :
 - a $G(\Lambda) \vdash \Rightarrow \varphi \rightarrow \chi$ and $G(\Lambda) \vdash \Rightarrow \chi \rightarrow \psi$;
 - b $\text{Prop}(\chi) \subseteq \text{Prop}(\varphi) \cap \text{Prop}(\psi)$;
 - c **$\text{Ag}(\chi) \subseteq \text{Ag}(\varphi) \cap \text{Ag}(\psi)$.**

Outline

- 1 Sequent Calculus $G(\mathbf{K45}_D)$
- 2 Analytic Cut Property of $G(\mathbf{K45}_D)$
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Syntax and Frame Classes

$$L \ni \varphi ::= p \mid \perp \mid \neg\varphi \mid \varphi \rightarrow \psi \mid D_G\varphi,$$

where $G \in \text{Grp}$ and $\text{Grp} := \{G \in \mathcal{P}(\text{Ag}) \mid G \neq \emptyset\}$.

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- $\mathbb{F}_{\mathbf{K45}}$: the class of all trans. and Euclidean frames.

Semantics

Let $M = (W, (R_a)_{a \in Ag}, V)$ be a model.

$M, w \models D_G \varphi$ iff for all $v \in W$,
 $(w, v) \in \bigcap_{a \in G} R_a$ implies $M, v \models \varphi$.

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$$\Box_a \varphi := D_{\{a\}} \varphi$$

Hilbert System

All the Axioms and Rules of H(K45_D)	
(Taut)	all instances of propositional tautologies
(Incl)	$D_G\varphi \rightarrow D_H\varphi \text{ } (G \subseteq H)$
(K)	$D_G(\varphi \rightarrow \psi) \rightarrow (D_G\varphi \rightarrow D_G\psi)$
(4)	$D_G\varphi \rightarrow D_GD_G\varphi$
(5)	$\neg D_G\varphi \rightarrow D_G\neg D_G\varphi$
(MP)	From $\varphi \rightarrow \psi$ and φ infer ψ
(Nec)	From φ infer $D_G\varphi$

Sequent Calculus

$$\varphi_1, \dots, \varphi_m \Rightarrow \psi_1, \dots, \psi_n$$

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$$\bigwedge \Gamma \text{ implies } \bigvee \Delta$$

Sequent Calculus

$$\frac{\Gamma \Rightarrow \Delta, \varphi \quad \varphi, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Cut)$$

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Sequent Calculus G(**K45**): ($\Rightarrow \Box_{\mathbf{K45}}$)

$$\frac{\varphi_1, \dots, \varphi_m, \Box \varphi_1, \dots, \Box \varphi_m \Rightarrow \Box \psi_1, \dots, \Box \psi_n, \chi}{\Box \varphi_1, \dots, \Box \varphi_m \Rightarrow \Box \psi_1, \dots, \Box \psi_n, \Box \chi}$$

Fact

$\Gamma \Rightarrow \Delta$ is derivable in G(**K45**) iff $\bigwedge \Gamma \rightarrow \bigvee \Delta$ is a theorem of H(**K45**).

Fact (Shvarts 1989)

The (*Cut*) rule is **admissible** in G(**K45**).

Sequent Calculus $G(\mathbf{K45}_D)$: $(\Rightarrow D^{\mathbf{K45}_D})$

$$\frac{\varphi_1, \dots, \varphi_m, D_{G_1}\varphi_1, \dots, D_{G_m}\varphi_m \Rightarrow D_{H_1}\psi_1, \dots, D_{H_n}\psi_n, \chi}{D_{G_1}\varphi_1, \dots, D_{G_m}\varphi_m \Rightarrow D_{H_1}\psi_1, \dots, D_{H_n}\psi_n, D_G\chi}$$

where $G_i, H_j \subseteq G$ for any $1 \leq i \leq m$ and $1 \leq j \leq n$.

Proposition

$\Gamma \Rightarrow \Delta$ is derivable in $G(\mathbf{K45}_D)$ iff $\bigwedge \Gamma \rightarrow \bigvee \Delta$ is a theorem of $H(\mathbf{K45}_D)$.

Proposition

The (*Cut*) rule is **not** admissible in $G(\mathbf{K45}_D)$.

$G(\mathbf{K45}_D)$ is Not Cut-free

$$\frac{\frac{D_{\{a\}}p \Rightarrow D_{\{a\}}p}{\Rightarrow \neg D_{\{a\}}p, D_{\{a\}}p} (\Rightarrow \neg) \quad \frac{p \Rightarrow p}{p, D_{\{a\}}p \Rightarrow p} (w \Rightarrow)}{\frac{\Rightarrow D_{\{a\}} \neg D_{\{a\}}p, D_{\{a\}}p}{\Rightarrow D_{\{a\}} \neg D_{\{a\}}p, D_{\{a,b\}}p} (\Rightarrow D^{\mathbf{K45}_D}) \quad \frac{D_{\{a\}}p \Rightarrow D_{\{a,b\}}p}{(Cut)_{D_{\{a\}}p}} (\Rightarrow D^{\mathbf{K45}_D})}$$

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This sequent is not derivable in $G(\mathbf{K45}_D)$ w/o (Cut) .

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$$\frac{}{\Rightarrow D_{\{a\}} \neg \textcolor{red}{D}_{\{a\}}p, D_{\{a,b\}}p} (Cut)_{D_{\{a\}}p}$$

Proposition

This sequent is not derivable in $G(\mathbf{K45}_D)$ w/o (Cut) .

Analytic Cut in $G(\mathbf{K45}_D)$

- A cut

$$\frac{\Gamma \Rightarrow \Delta, \varphi \quad \varphi, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Cut)_\varphi$$

is **analytic** if $\varphi \in \text{Sub}(\Gamma, \Pi, \Delta, \Sigma)$.

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- Define $G^a(\mathbf{K45}_D)$ as the same calculus as $G(\mathbf{K45}_D)$ except that **(Cut)** is always analytic.

(Thm.) Analytic Cut Property of $G(\mathbf{K45}_D)$

- If a sequent is derivable in $G(\mathbf{K45}_D)$,
then it is derivable in $G^a(\mathbf{K45}_D)$.
- If $\Gamma \Rightarrow \Delta$ is derivable in $G(\mathbf{K45}_D)$, then it is derivable in $G(\mathbf{K45}_D)$ by a derivation which consists of $\text{Sub}(\Gamma, \Delta)$ alone.

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(Cor.) Consistency of $G(\mathbf{K45}_D)$

The sequent $\Rightarrow$ is **not** derivable in $G(\mathbf{K45}_D)$.

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How to Establish Analytic Cut Property

If suffices to prove $(1) \Rightarrow (2) \Rightarrow (3)$:

- ① $\Gamma \Rightarrow \Delta$ is derivable in $G(\mathbf{K45}_D)$.
- ② $\Gamma \Rightarrow \Delta$ is valid in all $\mathbf{K45}_D$ pseudo frames $(W, (R_G)_{G \in \text{Grp}})$ where:
 - $G \subseteq H$ implies $R_H \subseteq R_G$;
 - $\bigcap_{a \in G} R_{\{a\}} = R_G$ may fail.
- ③ $\Gamma \Rightarrow \Delta$ is derivable in $G^a(\mathbf{K45}_D)$.
 - $(2) \Rightarrow (3)$ is semantic completeness of $G^a(\mathbf{K45}_D)$ for pseudo models.
 - A key notion is: $\exists$ -partial valuation due to Takano.

Ξ -partial Valuation

(Γ, Δ) is a Ξ -partial val. in $G^a(\mathbf{K45}_D)$ if:

- 1 $G^a(\mathbf{K45}_D) \not\models \Gamma \Rightarrow \Delta$;
- 2 $\text{Sub}(\Gamma, \Delta) = \Gamma \cup \Delta$; and
- 3 $\Gamma \cup \Delta \subseteq \Xi$.

Derived Pseudo Model From Ξ

Definition

$M_{\Xi} = (W_{\Xi}, (R_G)_{G \in \text{Grp}}, V_{\Xi})$ is defined as follows:

- W_{Ξ} is the set of all Ξ -partial valuations;
- $(\Gamma, \Delta) R_G (\Pi, \Sigma)$ iff $\forall H \subseteq G$

$(\{ \varphi, D_H \varphi \mid D_H \varphi \in \Gamma \} \subseteq \Pi \text{ and } \{ D_H \varphi \mid D_H \varphi \in \Pi \} \subseteq \Gamma);$

- $(\Gamma, \Delta) \in V_{\Xi}(p)$ iff $p \in \Gamma$.

Completeness

Theorem

If (2) $\Gamma \Rightarrow \Delta$ is valid in **all** $\mathbf{K45}_D$ pseudo frames, then (3) $\Gamma \Rightarrow \Delta$ is derivable in $G^a(\mathbf{K45}_D)$.

($\because$)

- 1 Suppose $G^a(\mathbf{K45}_D) \not\vdash \Gamma \Rightarrow \Delta$.
- 2 Let $\Xi := \text{Sub}(\Gamma, \Delta)$. There exists a Ξ -partial valuation (Γ^+, Δ^+) s.t. $\Gamma \subseteq \Gamma^+, \Delta \subseteq \Delta^+$.
- 3 Our goal $M_\Xi, (\Gamma^+, \Delta^+) \not\models \bigwedge \Gamma \rightarrow \bigvee \Delta$ is obtained by the definition of R_G .

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Craig Interpolation of $G(\mathbf{K45}_D)$

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Maehara's Method for a-Cut (Ono 1998)

By induction on proofs with **only** analytic applications of (*Cut*).

Adding Ag **does not** cause any trouble.

Conclusion

Main Result of This Talk

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- 2 (Corollary) CIP for $G(\mathbf{K45}_D)$, $G(\mathbf{KD45}_D)$ and $G(\mathbf{S5}_D)$.

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- 2 (Corollary) CIP for $G(\mathbf{K45}_D)$, $G(\mathbf{KD45}_D)$ and $G(\mathbf{S5}_D)$.
- 3 Adding the global modality $D_\emptyset$ **does not** change all the proof-theoretic results.

Further Directions

- 1 Sequent calculus for combination of coalition logic **CL** with distributed knowledge **CLD** (Ågotnes & Alchina 2016).
- 2 Syntactic proof for analytic cut property in **CLD** with parameter D_{Ag} alone.
- 3 Sequent calculus for expansions with comparative knowledge (Baltag & Smets 2020).